



Sino Bright School Chongqing

Midterm, 2025-2026 T2

Class & Subject: G10 A3 Mathematics

Date: Apr 2026

Time: 90 Minutes

Teacher: Shi Feng

IMPORTANT: Read together before beginning the exam.

Students must complete the assessments to the best of their ability. Any students found cheating during an exam will be given a zero and referred to the Principal for further disciplinary measures. Cheating is a serious offense.

“Cheating” may include any of the following:

- Having course papers, notes, unapproved data sheets or other course-related materials in the exam room without the expressed permission of the subject class teacher in charge of the exam.
- Using or even having a cellphone or other electronic device in the exam room without the express permission of the subject class teacher in charge of the exam.
- Attempting to communicate, either verbally or non-verbally with other students during the exam.
- Having exam-related materials prior to the writing of the exam.

Mark Breakdown:	Student Name: _____
Part I: Multiple Choice Questions 2 Marks $\times$ 10	
Part II: Short Answer Questions 2 Marks $\times$ 20	
Part III: Long Answer Questions 40 Marks	
Total Marks: 100 Marks	Score: _____

Special Instructions/Materials Allowed:

- Pen, Calculator

Multiple Choice Questions (10 questions, 2 marks each)

1. Simplify $\frac{x^2+5x+6}{x+2}$.
 - (a) $x + 3$
 - (b) $x - 3$
 - (c) $x + 2$
 - (d) $x - 2$
2. The remainder when $x^3 - 2x^2 + 4x - 5$ is divided by $(x - 1)$ is
 - (a) -2
 - (b) 2
 - (c) 4
 - (d) 0
3. Which of the following is a factor of $x^3 - 3x^2 - 4x + 12$?
 - (a) $x + 2$
 - (b) $x - 2$
 - (c) $x + 3$
 - (d) $x - 4$
4. The midpoint of the line segment joining $(-1, 4)$ and $(5, -2)$ is
 - (a) $(2, 1)$
 - (b) $(3, 1)$
 - (c) $(2, 3)$
 - (d) $(4, 2)$
5. The equation of the circle with centre $(2, -3)$ and radius 4 is
 - (a) $(x - 2)^2 + (y + 3)^2 = 4$
 - (b) $(x + 2)^2 + (y - 3)^2 = 16$
 - (c) $(x - 2)^2 + (y + 3)^2 = 16$
 - (d) $(x + 2)^2 + (y - 3)^2 = 4$
6. $\log_3 81$ is equal to
 - (a) 3
 - (b) 4
 - (c) 5
 - (d) 9
7. Solve $2^x = 16$.
 - (a) 2

- (b) 3
 - (c) 4
 - (d) 8
8. The discriminant of the quadratic $2x^2 - 4x + 5 = 0$ is
- (a) -24
 - (b) 24
 - (c) 56
 - (d) -56
9. The solutions to $x^2 - 5x + 6 = 0$ are
- (a) 2 and 3
 - (b) -2 and -3
 - (c) 1 and 6
 - (d) -1 and -6
10. The gradient of the line perpendicular to $y = 3x - 2$ is
- (a) 3
 - (b) -3
 - (c) $\frac{1}{3}$
 - (d) $-\frac{1}{3}$

True or False Questions (10 questions, 2 marks each)

State whether each statement is True or False.

- | | | |
|--|------|-------|
| 1. The factor theorem states that if $f(p) = 0$ then $(x - p)$ is a factor of $f(x)$. | True | False |
| 2. For the circle $(x - a)^2 + (y - b)^2 = r^2$, the centre is (a, b) and radius is r . | True | False |
| 3. $\log_a x + \log_a y = \log_a(x + y)$. | True | False |
| 4. The quadratic equation $x^2 + 4x + 5 = 0$ has two distinct real roots. | True | False |
| 5. The remainder theorem says that when $f(x)$ is divided by $(ax - b)$, the remainder is $f\left(\frac{b}{a}\right)$. | True | False |
| 6. The graph of $y = a^x$ with $a > 1$ is a decreasing function. | True | False |
| 7. The perpendicular bisector of a chord of a circle always passes through the centre. | True | False |
| 8. $\log_a a = 0$ for any $a > 0$, $a \neq 1$. | True | False |
| 9. Completing the square for $x^2 + 6x$ gives $(x + 3)^2 - 9$. | True | False |
| 10. The line $y = 2x + 1$ is a tangent to the circle $(x - 1)^2 + y^2 = 5$. | True | False |

Fill in the Blanks (10 questions, 2 marks each)

Complete each statement with the correct answer.

1. When simplifying the algebraic fraction $\frac{x^2 + 3x + 2}{x + 1}$, after cancelling common factors, the result is _____.
2. The remainder when $x^3 - 2x^2 + 4x - 5$ is divided by $(x - 2)$ is _____.
3. If $(x - 3)$ is a factor of $x^3 + ax^2 - 4x - 3$, then the value of a is _____.
4. The midpoint of the line segment joining $A(-3, 7)$ and $B(5, -1)$ is _____.
5. The equation of the circle with centre $(2, -5)$ and radius 3 is $(x - 2)^2 + (y + 5)^2 =$ _____.
6. The value of $\log_5 125$ is _____.
7. Solve $3^{2x-1} = 27$; then $x =$ _____.
8. The discriminant of the quadratic equation $2x^2 - 3x + 4 = 0$ is _____.
9. Completing the square for $x^2 + 8x$ gives $(x + 4)^2 -$ _____.
10. A circle has equation $x^2 + y^2 - 6x + 4y - 12 = 0$. The x -coordinate of its centre is _____.

Long Answer Questions

1. The polynomial $f(x) = 2x^3 - 5x^2 - 4x + 3$.

- (a) Use the factor theorem to show that $(x - 3)$ is a factor of $f(x)$. **(3 marks)**
- (b) Hence factorise $f(x)$ completely into linear factors. **(4 marks)**
- (c) Write down all the real roots of the equation $f(x) = 0$. **(3 marks)**

2. Consider the quadratic function $f(x) = x^2 - 6x + 5$.

- (a) Solve the equation $f(x) = 0$. **(2 marks)**
- (b) Write $f(x)$ in the form $(x - p)^2 + q$. **(3 marks)**
- (c) Hence state the minimum value of $f(x)$ and the value of x at which it occurs. **(2 marks)**
- (d) Find the discriminant of $f(x)$ and explain what it tells you about the roots. **(3 marks)**

3. The circle C has equation $(x - 1)^2 + (y + 4)^2 = 20$.

- (a) Write down the centre and radius of C . **(3 marks)**
- (b) Verify that the point $P(3, 0)$ lies on C . **(3 marks)**
- (c) Find the equation of the tangent to C at P . Give your answer in the form $ax + by + c = 0$, where a , b and c are integers. **(4 marks)**

4. (a) Solve $5^{2x-1} = 125$. **(3 marks)**
- (b) Solve $\log_2(x + 1) + \log_2(x - 1) = 3$. **(4 marks)**
- (c) Given that $\log_a 9 = 2$, find the value of a . **(3 marks)**